

Full Length Article

Novel uveal Melanocyte cell lines display distinct features from uveal melanoma

Lauren A. Dalvin^{a,b,c,d,*}, Piotr K. Kopinski^a, David R. Miley^a, Hien Ong^a,
Samantha A. Erickson^a, Cynthia M. Andrews-Pfannkoch^a, Stephanie D. Burr^a,
James A. Armitage^{c,d}

^a Departments of Ophthalmology, Mayo Clinic, Rochester, MN, USA

^b Medical Oncology, Mayo Clinic, Rochester, MN, USA

^c School of Medicine, Deakin University, Geelong, VIC, Australia

^d Institute for Mental and Physical Health and Clinical Translation (IMPACT), Deakin University, Geelong, VIC, Australia

ARTICLE INFO

Keywords:

Eye
Melanocyte
Melanoma
Uveal melanocyte
Uveal melanoma
Organoid
RNA-sequencing
Differentially expressed genes

ABSTRACT

Purposes: Despite advances in prognostication for uveal melanoma (UM), mortality from this disease remains high due to a lack of effective treatments to prevent metastatic progression. Novel therapeutic targets are needed, but few studies have been able to compare UM to its progenitor cell due to a lack of commercially available uveal melanocyte cell lines. We established novel uveal melanocyte cell lines for comparison to our novel UM patient-derived organoids (PDOs).

Methods: Uveal melanocytes were isolated from the choroid of post-mortem donor eyes. Melanocytes were cultured in Ham's F12 medium, and melanocytic markers were confirmed using immunohistochemistry. Melanocytes were grown as spheroids on Cultrex-coated plates prior to Exome- and RNA-sequencing for comparison to UM PDOs.

Results: Uveal melanocytes were successfully established with confirmed melanocytic markers and absence of contaminant cell types. Melanocytes displayed regular, higher order growth patterns in 2D culture compared to more primitive colony formation displayed by UM, and melanocytes could be carried for at least 15 passages. Exome-sequencing confirmed an absence of UM-relevant mutations in the melanocytes, and RNA-sequencing revealed differential gene expression between benign uveal melanocytes and UM PDOs.

Conclusions: Well-characterized uveal melanocyte cell lines can be compared to UM samples in the laboratory to help identify novel UM treatment targets and serve as a control population for future translation research.

1. Introduction

Uveal melanoma (UM) is the most common primary intraocular malignancy in adults, affecting up to 7095 new individuals across the world each year.^{1–3} UM remains a devastating disease.^{4,5} Standard of care treatment for primary UM is high-dose radiation, which often causes severe, irreversible vision loss.^{6–8} Furthermore, despite successful primary tumor treatment, up to 50% of patients will ultimately develop metastatic disease.⁹ Overall response rates for metastatic UM are < 20%,^{10–15} with median overall survival < 2 years.^{10,13} Unfortunately, extensive genomic sequencing efforts⁴ and advances in prognostication by UM subtypes^{4,16,17} have failed to identify mutations that can be targeted to treat UM. Translational research is urgently needed to improve outcomes for

patients diagnosed with this dismal disease.

A likely contributor to failed translation of historical UM research is a lack of laboratory models which accurately represent human UM subtypes. Many commercially available two-dimensional (2D) cell lines were derived from atypical primary UM tumors, and cell lines often deviate from the primary tumor in both chromosomal status and melanocyte markers.¹⁸ To overcome the shortcomings of these standard laboratory models, we developed and characterized a biobank of UM patient-derived organoids (PDOs), which better represent the broad molecular spectrum of human UM and retain key genetic markers and gene expression profiles of the primary tumor.¹⁹ Still, to most effectively study UM using these novel models, a well-validated control model is needed.

Uveal melanocytes, the presumed cell of origin for UM, remain

* Corresponding author. Departments of Ophthalmology, Mayo Clinic, 200 1st St SW, Rochester, MN, 55905, USA

E-mail address: dalvin.lauren@mayo.edu (L.A. Dalvin).

<https://doi.org/10.1016/j.aopr.2025.11.005>

Received 25 June 2025; Received in revised form 9 October 2025; Accepted 18 November 2025

Available online 19 November 2025

2667-3762/© 2025 The Authors. Published by Elsevier Inc. on behalf of Zhejiang University Press. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

understudied, with a scarcity of validated models.^{20,21} Uveal melanocyte cell lines are not commercially available, and in some cases, epidermal melanocytes have been used as control cells for UM studies,²² despite the known differences between UM and cutaneous melanoma.^{4,5} Herein, we describe development and characterization of novel uveal melanocyte cell lines along with comparison of features to UM models.

2. Methods

2.1. Establishing uveal melanocyte cell lines

Primary uveal melanocytes were isolated using modifications to a previously published protocol.²¹ Melanocytes were specifically isolated from the choroid to most closely approximate the origin cell for choroidal melanoma, which accounts for most cases of UM.²³ Briefly, donor eyes were obtained from Lion's Gift of Sight. The choroid was manually dissected and dissociated using the gentleMACS dissociator system (Miltenyi Biotect). Cells were resuspended and plated in Ham's F12 medium (Gibco) with 10% FBS, 50 µg/ml of gentamicin, 0.1 mM isobutylmethylxanthine, 10 ng/ml cholera toxin (Sigma), 20 ng/ml human recombinant growth factor (Stemcell), and 2 mM glutamine. Media was replaced twice per week, and cells were incubated at 37 °C in 5% CO₂ under humidified conditions. Cells were sent for short tandem repeat testing (Labcorp). Growth curves were assessed by seeding 3×10^5 cells on three 6-well plates, with each replicate consisting of a full row of a plate. Adherent cells were counted with media changes every 4 days. Growth curves were compared to those seen with two commercially available uveal melanoma cell lines, MP46 and MP65 (ATCC CRL-3298 and CRL-3299, respectively).

2.2. Immunohistochemistry

Immunohistochemistry (IHC) was done as described to validate the melanocytic nature of the cell lines.^{19,24,25} The following primary antibodies were evaluated: Melan-A 1:200 (Abcam ab785), SOX10 1:200 (Abcam ab227680), and Bestrophin 1 (NOVUS NB300-164). The following secondary antibodies were used: Alexa Fluor 647 (Abcam ab150083) and Alexa Fluor 488 (Abcam ab150113). Vectashield Plus containing DAPI (Vector Laboratories H-2000) was used for nuclear staining. Images were analyzed with a confocal laser microscope (Zeiss LSM 780).

2.3. Exome- and RNA-sequencing

Exome- and RNA-sequencing were done as previously described.^{19,26–28} Prior to DNA or RNA extraction for sequencing studies, uveal melanocytes were grown as three-dimensional (3D) spheroids via plating on Cultrex-coated plates to permit optimal comparison with our

published UM PDOs.¹⁹

Briefly, for exome-sequencing after library preparation and sequencing, raw BCL files were converted to FASTQ, and data quality control (QC) and analysis were performed using DNA-seq Analysis Pipeline. Clean reads were aligned to the GRCh38 reference genome using Sentieon 202112.01. Somatic SNVs and small INDELs were called by using Sentieon 202112.01 (TNseq algorithm). The VCF files were normalized (left alignment of INDELs and splitting multiallelic sites into multiple sites) using bcftools 1.13. Overlapped transcripts were identified for each variant, and the effects of variants on the transcripts were predicted by Ensembl VEP 104. Impact of the variants was classified using the R package for MAfTools 2.14.0.

RNA-sequencing was done as previously described by Azenta Life Sciences using the Illumina Ultra-Low Input protocol for library preparation and Illumina 2 × 150 bp PE reads.¹⁹ RNA-sequencing FASTQ files were aligned to HG38 Reference Genome using STAR 2.7.8a. Gene-level counts were obtained using HTseq-count v.0.9.1 (Galaxy), and differential gene expression analysis was done with R package DESeq2 version 2.11.40.7 (RStudio). Genes with a base mean ≥ 1 , $\log_2(\text{fold change, FC}) > 1$ or < -1 , and an adjusted *P* value < 0.05 were considered significantly differentially expressed. Heatmaps were created using the R package ggplot2 version 3.3.3. GraphPad Prism version 10.5.0 was used to visualize heatmaps and PCA plots. Ingenuity Pathway Analysis (IPA, Qiagen) was used to identify upregulated and downregulated pathways.

2.4. Data availability

Sequencing data described in this report have been deposited in Sequence Read Archive (BioProject accession number PRJNA1068520). Sequencing was done at passage 1, with confirmation of stability at passage 3 for a subset samples.

3. Results

3.1. Primary uveal melanocytes can be successfully established from donor eyes

We attempted to generate uveal melanocyte cell lines from bilateral eyes of 3 unique donor patients. All donor patients were white, matching the demographics of patients who contributed samples to our published UM PDO biobank.¹⁹ The first two cell lines (PK1, PK2) were cultured without antibiotics and appeared to contain bacterial contamination. The remaining cell lines were cultured with gentamicin. Of those, one cell line (PK6) appeared contaminated with another cell type. The remaining cell lines (PK3, PK4, PK5) contained successfully isolated populations of uveal melanocytes, and STR testing supported the novel nature of the cell lines (Supplemental Fig. 1). Melanocytic origin of the

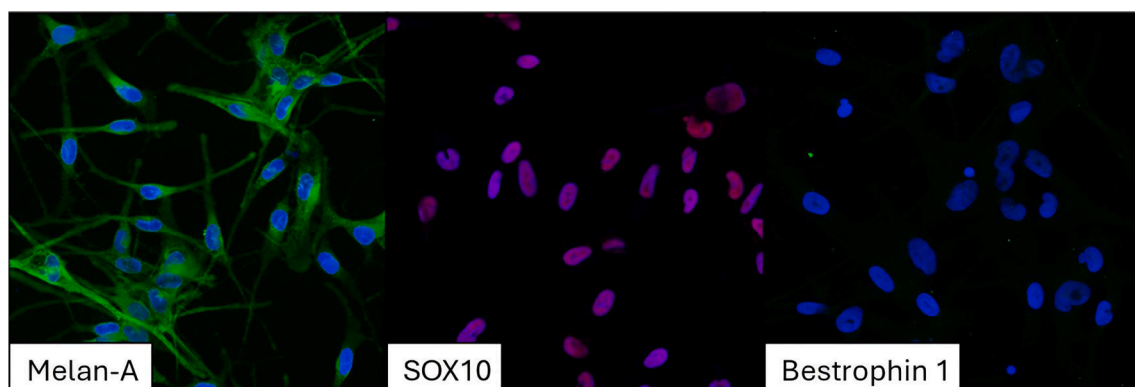


Fig. 1. Uveal melanocyte cell lines display appropriate melanocytic markers.

Uveal melanocytes displayed expected immunohistochemical staining for Melan-A and SOX10 and absent staining for the retinal pigment epithelial cell marker Bestrophin 1. Results were consistent throughout passing with uniform staining of all cells. Representative images are shown.

cells was confirmed by IHC for Melan-A and SOX10, with absence of retinal pigment epithelial (RPE) cell contamination confirmed by negative Bestrophin 1 staining (Fig. 1). In 2D culture, uveal melanocytes exhibited a honeycomb-like growth pattern, with regular, alternating areas of cellular density and paucity. In contrast, UM in 2D culture formed irregular colonies with lack of clear patterns or contact inhibition (Fig. 2). At the time of this writing, uveal melanocyte cell lines continue to grow up to passage 15. Growth curves for PK3 and PK4 compared to uveal melanoma cell lines MP46 and MP65 demonstrated moderate growth rate of benign melanocytes, in between the growth

rates for the two malignant cell lines (Fig. 3).

3.2. Exome-sequencing confirms absence of UM mutations in melanocyte cell lines

We grew PK3, PK4, and PK5 as 3D spheroids plated on Cultrex and sent them for exome-sequencing batched with a cohort of our PDOs. Sequencing confirmed UM mutations matching donor primary tumors in UM PDOs as previously published¹⁹ and demonstrated an absence of UM pathogenic mutations in benign melanocyte cell lines (Fig. 4). PK3 and

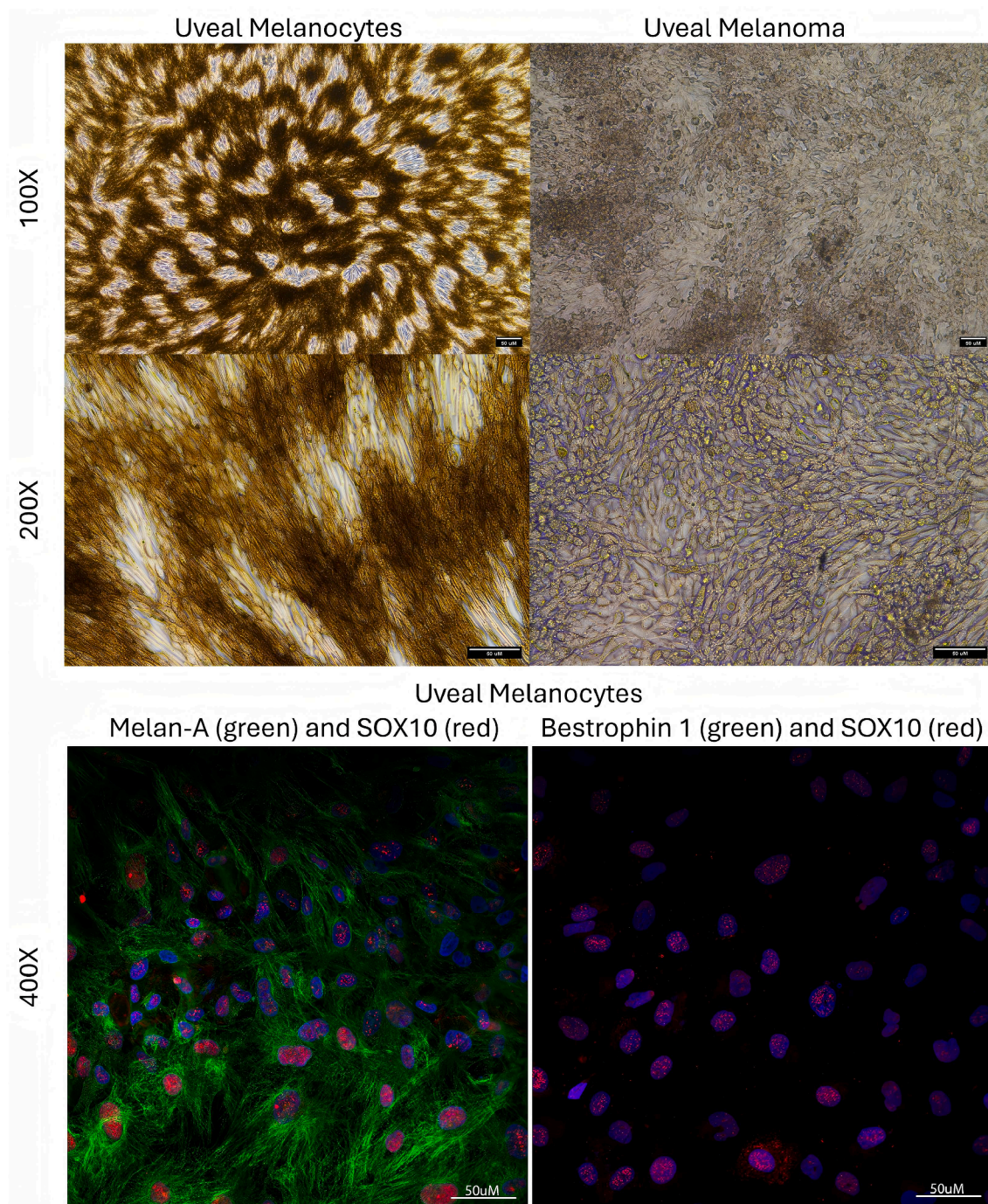


Fig. 2. Uveal melanocytes displayed higher order growth patterns than uveal melanoma.

In two-dimensional cell culture, uveal melanocytes formed regular, honeycomb-like growth patterns, with repeating areas of cellular density and cellular paucity. Uveal melanoma cells, in contrast, formed more primitive colonies and displayed lack of contact inhibition. A representative image is shown of uveal melanocytes at passage 3. Immunohistochemical staining of the melanocytes shows all cells are positive for Melan-A and SOX10 and negative for Bestrophin 1.

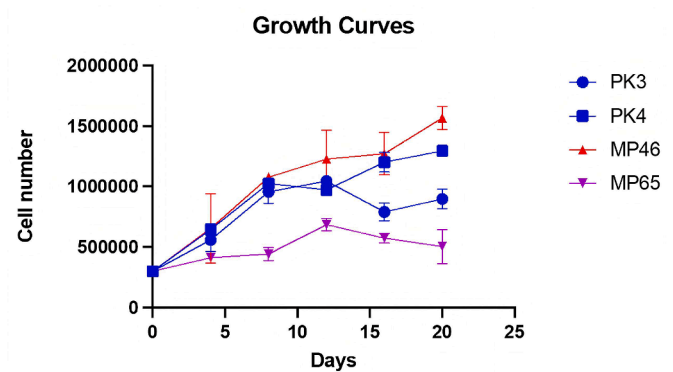


Fig. 3. Uveal melanocytes displayed moderate growth rate compared to uveal melanoma. Growth curves of benign uveal melanocytes PK3 and PK4 at passage 1 compared to commercially available uveal melanoma cell lines MP46 and MP65 show intermediate growth rates of melanocytes compared to melanoma.

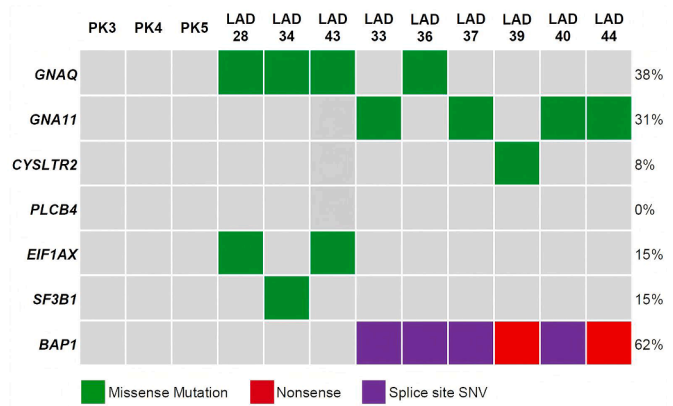
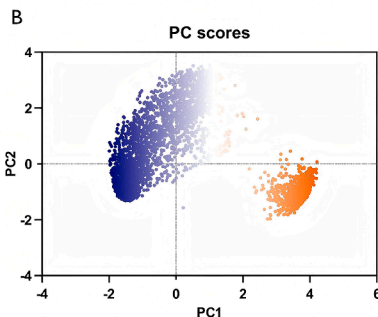
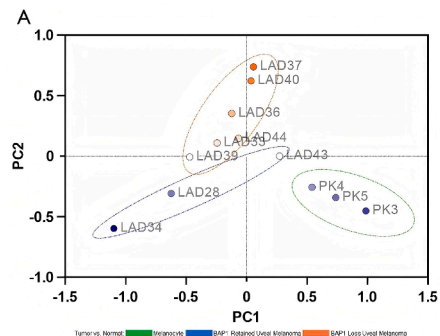


Fig. 4. Uveal melanocytes were free from pathogenic mutations relevant to uveal melanoma. Oncoplot shows relevant uveal melanoma mutations in uveal melanoma patient-derived organoids (LADs) and absence of any such mutations in uveal melanocytes (PKs). Of note, PK3 and PK4 contained the same variant, rs123598, in the 3'UTR of BAP1, which is reported to be a benign variant. Sequencing was done at passage 1 and confirmed at passage 3.

PK4 contained the same variant, rs123598, in the 3'UTR of BAP1, which is likely a benign variant given its reporting in the 1000 Genomes Project with a Caucasian minor allele frequency (MAF) of 0.06 (A) and a global MAF of 0.03.²⁹



Tumors: Normal (green), Melanocyte (blue), BAP1 Retained Uveal Melanoma (orange), BAP1 Loss Uveal Melanoma (red)

3.3. Uveal melanocytes display differential gene expression from UM

Passage 1 PK3, PK4, and PK5 were grown as 3D spheroids to match the 3D environment of our UM PDOs prior to sending passage 1 3D spheroid-derived material for RNA-sequencing. PCA plots are shown, demonstrating close clustering of all three melanocyte cultures with a broader spread of BAP1 retained and BAP1 loss UM PDOs, representing UM subtypes at low (BAP1 retained) or high (BAP1 loss) risk for progression to metastasis^{30,31} (Fig. 5). To provide clinical correlative information on outcomes from PDO donor patients, details on metastatic progression and survival are provided in Table 1.

Comparing the three uveal melanocyte cultures to a cohort of 9 UM PDOs (n = 3 with BAP1 retained, n = 6 with BAP1 loss), there were 4505 differentially expressed genes (DEGs) of which 4505 had at least 1-fold change between groups. Of the 4505 significant DEGs with at least 1-fold change, 3630 DEGs were upregulated in UM compared to melanocytes, while 875 DEGs were downregulated in UM. The top 50 upregulated and downregulated DEGs are shown in Fig. 6, which shows the top 50 DEGs excluding uncharacterized genes denoted by LOC, and Supplementary Table 1, which shows the top 50 DEGs including uncharacterized genes.

Compared to one prior publication which separately investigated DEGs in normal uveal melanocytes versus two commercial uveal melanoma cell lines (MP41 and MP46), we found significant up- or downregulation in 21 (42%) of the previously reported top 50 DEGs (Supplementary Table 2).³² However, only 4 of the previously reported top 50 DEGs (Up in UM: *HAGLR*, *ASB11*, *TRPM2*; Down in UM; *GFRA3*)³² were present in either our top 50 upregulated DEGs or top 50 downregulated DEGs. These results indicate potential importance of these genes but also the possibility of uncovering novel DEGs due to our inclusion of a broader spectrum of UM PDOs which better represent diverse human UM subtypes.¹⁹

Table 1
Clinical outcomes for donor patients for uveal melanoma patient-derived organoids (PDOs).

PDO	BAP1 Expression	Metastasis	Time to metastasis (months)	Follow-up (months)	Status at Last Follow-up
LAD28	retained	No		41	Alive
LAD34	retained	No		38	Alive
LAD43	retained	No		18	Dead, unrelated cause
LAD33	lost	No		38	Alive
LAD36	lost	Yes	6	32	Dead from melanoma
LAD37	lost	Yes	20	35	Alive
LAD39	lost	Yes	7	12	Alive
LAD40	lost	No		33	Alive
LAD44	lost	Yes	15	31	Alive

Fig. 5. Uveal melanocytes have differential gene expression compared to uveal melanoma. (A) PCA plot and (B) PCA scores clustered by groups of uveal melanocytes (PKs) or uveal melanoma (LADs). On the PCA plot, three clusters could be visualized representing benign uveal melanocytes (green), BAP1 retained uveal melanoma (blue), and BAP1 loss uveal melanoma (orange). While all three benign melanocyte cultures clustered tightly together, there was a broader spread of malignant uveal melanoma patient-derived organoids, which clustered into BAP1 retained and BAP1 loss subtypes. These represent uveal melanoma subtypes that are at low (BAP1 retained) or high (BAP1 loss) risk for progression to metastasis. Dot fill color corresponds with the PC color scale to the right of the plot.

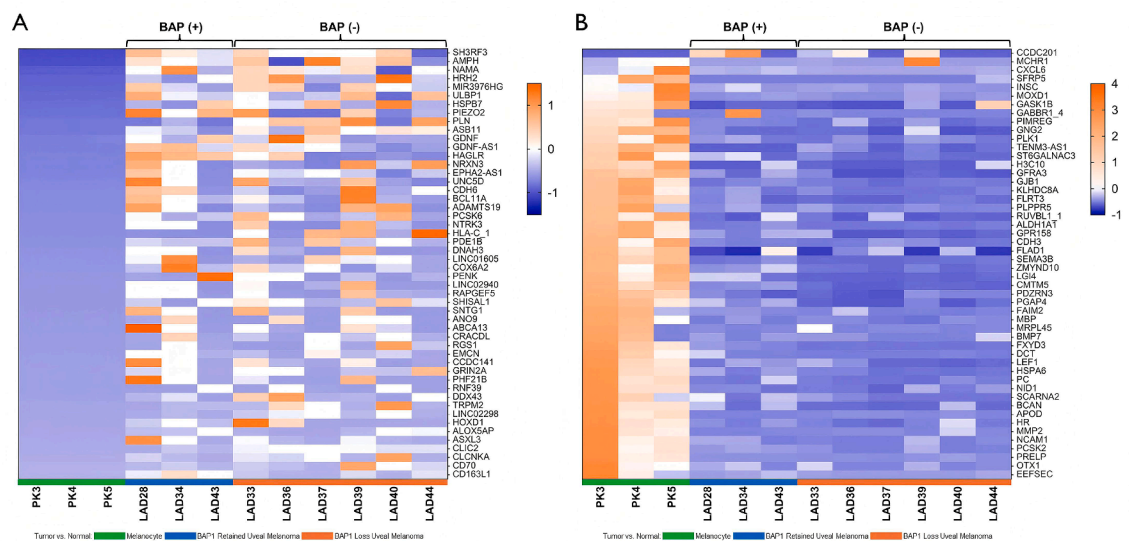


Fig. 6. Uveal melanoma displays differential expression of oncogenically-relevant genes compared to benign uveal melanocytes. Heat maps show the top 50 differentially expressed genes (DEGs) that were (A) upregulated or (B) downregulated in uveal melanoma patient-derived organoids compared to benign uveal melanocytes grown as three-dimensional spheroids.

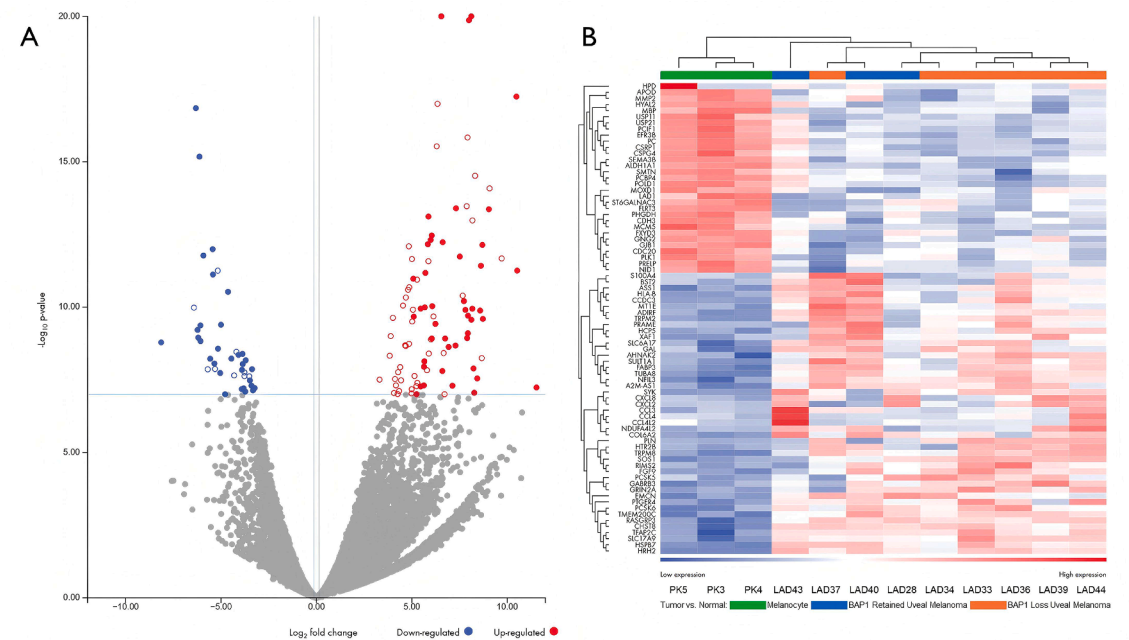


Fig. 7. Uveal melanoma displays upregulation of genes related to cancer functions compared to benign uveal melanocytes. (A) Volcano plot shows differentially expressed genes (DEGs) filtered for $P \leq 1 \times 10^{-7}$ with corresponding (B) heat map showing DEGs in uveal melanoma tumor (LADs) versus normal benign melanocytes (PKs). When visualizing the top up- and downregulated pathways related to disease and functions in benign uveal melanocytes compared to uveal melanoma patient-derived organoids, cancer functions were among the most upregulated in uveal melanoma.

We visualized the top up- and downregulated pathways related to disease and functions between uveal melanocytes and UM PDOs. Not surprisingly, cancer functions were among the most upregulated in UM (Fig. 7).

4. Discussion

Translational research is needed to uncover novel strategies for targeting UM, a dismal disease in which about half of patients will ultimately develop incurable metastasis.^{9–15} Unfortunately, shortcomings in standard in vitro UM models have limited the successful translation of laboratory studies. Standard models have often been derived from

atypical tumors and may have drifted in chromosomal status or melanocyte markers compared to the corresponding primary tumor.^{18,33,34} Our laboratory previously took initial steps to address these shortcomings by developing novel UM PDOs, which better represent the spectrum of human UM subtypes, genotypically and phenotypically.¹⁹ Still, a reference in vitro control population is warranted to facilitate discovery of novel targetable pathways unique to UM and investigate drug specificity for cancerous versus benign cell populations. Benign uveal melanocytes, the precursor cell for UM, are not available for commercial purchase, and there are limited reports to date comparing features of UM with uveal melanocytes.^{20,21,32} In this work, we expanded our biobank of models by generating three novel benign uveal melanocyte cell lines.

We validated the melanocytic and benign nature of these cell lines, grew them as 3D spheroids to serve as controls for our UM PDOs, and probed them to identify DEGs compared to a diverse array of UM subtypes represented in our PDO biobank.¹⁹

There were several genes of interest upregulated in UM. *PCSK6*, also known as *PACE4*, was in the top 10 upregulated genes in our dataset; its expression has been detected in human primary and metastatic cutaneous melanoma, while it was absent in situ disease.³⁵ Increased *PACE4* expression enhanced in vivo cutaneous melanoma growth in a murine model, and in vitro cell proliferation, MMP-2 production, gelatinase activity, and migratory capacity were increased.³⁵ Our data suggest that investigation may be warranted in UM. *GRIN2A* was another upregulated gene. *GRIN2A* mutations are frequent in cutaneous melanoma, and certain mutations increase oncogenesis in melanoma.^{36–38} *RGS1* encodes a member of the regulator of G protein family and was upregulated in UM. The role of elevated *RGS1* has been previously described in cutaneous melanoma, where it is associated with increased tumor thickness, mitotic rate, vascular involvement, sentinel lymph node metastasis, and reduced relapse-free and disease-specific survival.³⁹ In the lab, *RGS1* has been associated with proliferation and migration of cutaneous melanoma cells, and it may promote melanoma progression through downstream signaling via G protein pathways to activate AKT and ERK.⁴⁰ Such pathways are especially relevant to UM, as 90% or more of UM is driven by activating mutations in *GNAQ* or *GNA11*.^{4,41} Although *RNF39* had the largest log₂FC (22.4) of all DEGs upregulated in UM, this seemed artificially elevated given the high standard error (3.53) and the next largest log₂FC of 11.9. Nevertheless, *RNF39* was upregulated in cutaneous melanoma compared to normal skin tissue⁴² and was differentially expressed between primary cutaneous melanoma and cutaneous melanoma metastasis⁴³; this may warrant further investigation in UM.

Regarding downregulated genes, one of the most downregulated in UM was *PRELP*. *PRELP* expression is reduced in cutaneous melanoma compared to adjacent normal tissue and has been associated with reduced survival.⁴⁴ In cutaneous melanoma, *PRELP* functions to induce MHC class I and CCL5 expression which may enhance T cell recruitment and immune response against the cancer.⁴⁴ *APOD* may have a unique function in UM, as its expression was downregulated, while elevated *APOD* has been associated with worse disease in other cancers such as cutaneous melanoma⁴⁵ and gastric cancer.⁴⁶ Similarly, *ALDH1A1* is downregulated in our UM samples compared to melanocytes, but the cutaneous melanoma literature has described that overexpression of *ALDH1A1* promotes tumor angiogenesis,⁴⁷ increases tumorigenesis,⁴⁸ and confers resistance to BRAF and MEK inhibitors.⁴⁹ These findings highlight the importance of studying UM as a unique disease, which is distinct from its cutaneous counterpart.

A few key DEGs were common in both our work and prior work by Gentien et al.³² Upregulated genes included *HAGLR*, *ASB11*, and *TRPM2*. *HAGLR* and *ASB11* overexpression have been demonstrated in cutaneous melanoma, with suppression of viability and cell migration following either *HAGLR* or *ASB11* knockdown.⁵⁰ Knockdown of the long non-coding RNA (lncRNA) *HAGLR* inhibited viability and motility of pancreatic cancer.⁵¹ *TRPM2* expression is also increased in cutaneous melanoma, with a role in growth, proliferation, and survival, demonstrating potential utility of *TRPM2* inhibition as a therapeutic target.^{52,53} Given the lack of consistency found in downregulated genes between the two commercial UM cell lines (vs. melanocytes) analyzed by Gentien et al.,³² it is not surprising that our downregulated genes had little overlap with their findings. Only *GFRA3* was in the top 50 downregulated DEGs in UM in both our data and that of Gentien et al.³² *GFRA3* promoter hypermethylation was associated with worse post-operative survival in gastric cancer,⁵⁴ but the role of *GFRA3* in melanoma is less well-described. Although we found that many genes were consistent with the findings by Gentien et al.,³² it's important to note that consistent upregulation or downregulation was not seen across all our UM PDOs. Gentien et al. explored only two UM samples, MP41

and MP46, both commercial cell lines. MP41 is BAP1 retained, which is a UM subtype that less often leads to metastasis.^{18,32} However, metastasis did develop in the patient who donated tissue for MP41. This sample may be less representative of most patients who will develop metastatic UM. MP46 is also somewhat unusual in that it has loss of BAP1 expression but no detected mutation in the coding region for BAP1⁵⁵; rather, the BAP1 promoter is deleted in MP46 with hyper-methylation of the deletion boundaries.³² Considering a future utilizing personalized medicine strategies for UM, it will be important to examine UM subtype-specific targets in a more diverse cohort of samples representing the spectrum of typical human UM such as those in our PDO biobank.

In summary, we generated three novel benign uveal melanocyte cell lines for comparison with UM. We successfully passaged our uveal melanocytes through passage 15, permitting their ongoing use as control cell populations for mechanistic and functional experiments. In these studies, we compared DEGs in melanocytes to a broad molecular spectrum of UM using PDO models that better represent human disease than commercial cell lines.¹⁹ We found some DEGs consistent with those in a prior commercial cell line study³² but also uncovered some distinct potential targets. Importantly, we demonstrated that gene expression can vary across a cohort of PDOs representing distinct UM subtypes, highlighting the importance of studying multiple models to propel personalized medicine. Future studies could investigate additional multi-omics comparison of melanocytes to a cohort of PDOs and investigate the UM-specific function of candidate targets uncovered in this work.

Study approval

Collection of uveal melanocytes from donor eyes was done in a deidentified fashion following collection of consent for donation of eye tissue from the Lions Eye Bank. Collection of material to develop patient-derived uveal melanoma organoids was approved by the IRB at Mayo Clinic Rochester with IRB number 19–000946.

Author contributions

The authors confirm contribution to the paper as follows: LAD: Conceptualization, Investigation, Writing-original draft, Supervision. PKK: Conceptualization, Investigation, DRM, HO: Investigation, Writing-review & editing. SAE, CMAP: Investigation. SDB: Investigation, Writing-review & editing. JAA: Investigation, Writing-review & editing, Supervision. All authors read and approved the submitted version.

Funding

This research was supported by the Center for Biomedical Discovery, Mayo Clinic, Rochester, MN, the Bruce A. Chambers Career Development Award in Cancer Honoring David Dingli, M.D., Ph.D., the Department of Oncology Small Grants Program, the National Center for Advancing Translational Sciences (NCATS) [grant numbers UL1TR002377 and KL2 TR002379], and the National Cancer Institute [grant number P30 CA015083].

Declaration of competing interest

Dr. Dalvin is a Consultant for Aura Biosciences and IDEAYA Biosciences. Neither company had any role in the design or publication of this work. No other authors had any conflicts of interest.

Acknowledgements

Thanks to Lions Gift of Sight for procurement of donor eyes for research.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.aopr.2025.11.005>.

References

- Singh AD, Turell ME, Topham AK. Uveal melanoma: trends in incidence, treatment, and survival. *Ophthalmology*. 2011;118(9):1881–1885. <https://doi.org/10.1016/j.ophtha.2011.01.040>.
- Kivelä T. The epidemiological challenge of the Most frequent eye cancer: retinoblastoma, an issue of birth and death. *Br J Ophthalmol*. 2009;93(9):1129–1131. <https://doi.org/10.1136/bjo.2008.150292>.
- Beasley AB, Preen DB, McLennan S, et al. Incidence and mortality of uveal melanoma in Australia (1982–2014). *Br J Ophthalmol*. 2023;107(3):406–411. <https://doi.org/10.1136/bjophthalmol-2021-319700>.
- Robertson AG, Shih J, Yau C, et al. Integrative analysis identifies four molecular and clinical subsets in uveal melanoma. *Cancer Cell*. 2017;32(2). <https://doi.org/10.1016/j.ccell.2017.07.003>, 204–20.e15.
- Genomic classification of cutaneous melanoma. *Cell*. 2015;161(7):1681–1696. <https://doi.org/10.1016/j.cell.2015.05.044>.
- Dalvin LA, Zhang Q, Hamerschock RA, et al. Nomogram for visual acuity outcome after iodine-125 plaque radiotherapy and prophylactic intravitreal bevacizumab for uveal melanoma in 1131 patients. *Br J Ophthalmol*. 2020;104(5):697–702. <https://doi.org/10.1136/bjophthalmol-2019-314686>.
- Melia BM, Abramson DH, Albert DM, et al. Collaborative ocular melanoma study (COMS) randomized trial of I-125 brachytherapy for medium choroidal melanoma. I. Visual acuity after 3 years COMS report no. 16. *Ophthalmology*. 2001;108(2):348–366. [https://doi.org/10.1016/s0161-6420\(00\)00526-1](https://doi.org/10.1016/s0161-6420(00)00526-1).
- Shields CL, Dalvin LA, Chang M, et al. Visual outcome at 4 years following plaque radiotherapy and prophylactic intravitreal bevacizumab (every 4 months for 2 years) for uveal melanoma: Comparison with nonrandomized historical control individuals. *JAMA Ophthalmol*. 2020;138(2):136–146. <https://doi.org/10.1001/jamaophthalmol.2019.5132>.
- Xu TT, Moser JC, Dalvin LA. Uveal melanoma: laboratory advances and new frontiers in patient care. *Curr Opin Ophthalmol*. 2021;32(3):301–308. <https://doi.org/10.1097/icu.0000000000000744>.
- Rantala ES, Hernberg M, Kivelä TT. Overall survival after treatment for metastatic uveal melanoma: a systematic review and meta-analysis. *Melanoma Res*. 2019;29(6):561–568. <https://doi.org/10.1097/cmr.0000000000000575>.
- Orloff M, Carvajal RD, Shoushtari AN, et al. Overall survival in patients who received checkpoint inhibitors after completing tebentafusp in a phase 3 randomized trial of first-line metastatic uveal melanoma. *J Clin Oncol*. 2021;39(15 suppl):9526. https://doi.org/10.1200/JCO.2021.39.15_suppl.9526.
- Piperno-Neumann S, Hassel JC, Rutkowski P, et al. Abstract CT002: phase 3 randomized trial comparing tebentafusp with investigator's choice in first line metastatic uveal melanoma. *Cancer Res*. 2021;81(13 Supplement):CT002-CT. <https://doi.org/10.1158/1538-7445.Am2021-ct002>.
- Nathan P, Hassel JC, Rutkowski P, et al. Overall survival benefit with tebentafusp in metastatic uveal melanoma. *N Engl J Med*. 2021;385(13):1196–1206. <https://doi.org/10.1056/NEJMoa2103485>.
- Olofsson Bagge R, Nelson A, Shafazand A, et al. Isolated hepatic perfusion with melphalan for patients with isolated uveal melanoma liver metastases: a multicenter, randomized, open-label, phase III trial (the SCANDIUM trial). *J Clin Oncol*. 2023;41(16):3042–3050. <https://doi.org/10.1200/jco.2022.01705>.
- Szeligo BM, Ivey AD, Boone BA. Poor response to checkpoint immunotherapy in uveal melanoma highlights the persistent need for innovative regional therapy approaches to manage liver metastases. *Cancers (Basel)*. 2021;13(14). <https://doi.org/10.3390/cancers13143426>.
- Harbour JW. A prognostic test to predict the risk of metastasis in uveal melanoma based on a 15-gene expression profile. *Methods Mol Biol*. 2014;1102:427–440. https://doi.org/10.1007/978-1-62703-727-3_22.
- Jager MJ, Brouwer NJ, Esmaili B. The cancer genome atlas project: an integrated molecular view of uveal melanoma. *Ophthalmology*. 2018;125(8):1139–1142. <https://doi.org/10.1016/j.ophtha.2018.03.011>.
- Jager MJ, Magner JA, Ksander BR, Dubovy SR. Uveal melanoma cell lines: where do they come from? (an American ophthalmological Society Thesis). *Trans Am Ophthalmol Soc*. 2016;114:T5.
- Dalvin LA, Andrews-Pfannkock CM, Miley DR, et al. Novel uveal melanoma patient-derived organoid models recapitulate human disease to support translational research. *Investig Ophthalmol Vis Sci*. 2024;65(13):60. <https://doi.org/10.1167/iov.65.13.60>.
- Hu F, Teramura DJ, Mah K. Normal uveal melanocytes in culture. *Pigment Cell Res*. 1987;1(2):94–103. <https://doi.org/10.1111/j.1600-0749.1987.tb00396.x>.
- Hu DN, McCormick SA, Ritch R, Pelton-Henrion K. Studies of human uveal melanocytes in vitro: isolation, purification and cultivation of human uveal melanocytes. *Investig Ophthalmol Vis Sci*. 1993;34(7):2210–2219.
- Rago F, Elliott G, Li A, et al. The discovery of SWI/SNF chromatin remodeling activity as a novel and targetable dependency in uveal melanoma. *Mol Cancer Therapeut*. 2020;19(10):2186–2195. <https://doi.org/10.1158/1535-7163.Mct-19-1013>.
- Shields CL, Kaliki S, Furuta M, et al. Clinical spectrum and prognosis of uveal melanoma based on age at presentation in 8,033 cases. *Retina*. 2012;32(7):1363–1372. <https://doi.org/10.1097/IAE.0b013e31824d09a8>.
- Chowdhury UR, Bahler CK, Hann CR, et al. ATP-sensitive potassium (KATP) channel activation decreases intraocular pressure in the anterior chamber of the eye. *Investig Ophthalmol Vis Sci*. 2011;52(9):6435–6442. <https://doi.org/10.1167/iov.11-7523>.
- Dalvin LA, Fautsch MP. Analysis of circadian rhythm gene expression with reference to diurnal pattern of intraocular pressure in mice. *Investig Ophthalmol Vis Sci*. 2015;56(4):2657–2663. <https://doi.org/10.1167/iov.15-16449>.
- Lenkiewicz E, Malasi S, Hogenson TL, et al. Genomic and epigenomic landscaping defines new therapeutic targets for adenocarcinoma of the pancreas. *Cancer Res*. 2020;80(20):4324–4334. <https://doi.org/10.1158/0008-5472.Can-20-0078>.
- Hogenson TL, Xie H, Phillips WJ, et al. Culture media composition influences patient-derived organoid ability to predict therapeutic responses in gastrointestinal cancers. *JCI Insight*. 2022;7(22). <https://doi.org/10.1172/jci.insight.158060>.
- Robinson JT, Thorvaldsdóttir H, Winckler W, et al. Integrative genomics viewer. *Nat Biotechnol*. 2011;29(1):24–26. <https://doi.org/10.1038/nbt.1754>.
- Rai K, Pilarski R, Boru G, et al. Germline BAP1 alterations in familial uveal melanoma. *Genes Chromosomes Cancer*. 2017;56(2):168–174. <https://doi.org/10.1002/gcc.22424>.
- Szalai E, Wells JR, Ward L, Grossniklaus HE. Uveal Melanoma nuclear BRCA1-Associated Protein-1 immunoreactivity is an Indicator of metastasis. *Ophthalmology*. 2018;125(2):203–209. <https://doi.org/10.1016/j.ophtha.2017.07.018>.
- Ida CM, Pulido J, Greipp PT, et al. BAP1 immunostain status in intraocular Biopsy specimens for uveal Melanoma highly correlates with other prognostic markers. *Ocul Oncol Pathol*. 2022;8(1):22–29. <https://doi.org/10.1159/000515858>.
- Gentien D, Saberi-Ansari E, Servant N, et al. Multi-omics comparison of malignant and normal uveal melanocytes reveals molecular features of uveal melanoma. *Cell Rep*. 2023;42(9):113132. <https://doi.org/10.1016/j.celrep.2023.113132>.
- Amirouchene-Angelozzi N, Nemati F, Gentien D, et al. Establishment of novel cell lines recapitulating the genetic landscape of uveal melanoma and preclinical validation of mTOR as a therapeutic target. *Mol Oncol*. 2014;8(8):1508–1520. <https://doi.org/10.1016/j.molonc.2014.06.004>.
- Némati F, Sastre-Garau X, Laurent C, et al. Establishment and characterization of a panel of human uveal melanoma xenografts derived from primary and/or metastatic tumors. *Clin Cancer Res*. 2010;16(8):2352–2362. <https://doi.org/10.1158/1078-0432.Ccr-09-3066>.
- Weishaupt C, Mastrofrancesco A, Metz D, et al. Paired basic amino acid-cleaving enzyme 4 (PCSK6): an emerging new target molecule in Human Melanoma. *Acta Derm Venereol*. 2020;100(10). <https://doi.org/10.2340/00015555-3525>, adv00157.
- D'Mello SA, Joseph WR, Green TN, et al. Selected GRIN2A mutations in melanoma cause oncogenic effects that can be modulated by extracellular glutamate. *Cell Calcium*. 2016;60(6):384–395. <https://doi.org/10.1016/j.ceca.2016.09.003>.
- Prickett TD, Zerlanko BJ, Hill VK, et al. Somatic mutation of GRIN2A in malignant melanoma results in loss of tumor suppressor activity via aberrant NMDAR complex formation. *J Invest Dermatol*. 2014;134(9):2390–2398. <https://doi.org/10.1038/jid.2014.190>.
- Wei X, Walia V, Lin JC, et al. Exome sequencing identifies GRIN2A as frequently mutated in melanoma. *Nat Genet*. 2011;43(5):442–446. <https://doi.org/10.1038/ng.810>.
- Rangel J, Nosrati M, Leong SP, et al. Novel role for RGS1 in melanoma progression. *Am J Surg Pathol*. 2008;32(8):1207–1212. <https://doi.org/10.1097/PAS.0b013e31816f53c>.
- Sun MY, Wang Y, Zhu J, et al. Critical role for non-GAP function of Gαs in RGS1-mediated promotion of melanoma progression through AKT and ERK phosphorylation. *Oncol Rep*. 2018;39(6):2673–2680. <https://doi.org/10.3892/or.2018.6341>.
- Jager MJ, Shields CL, Cebulla CM, et al. Uveal melanoma. *Nat Rev Dis Primers*. 2020;6(1):24. <https://doi.org/10.1038/s41572-020-0158-0>.
- Han Y, Li X, Yan J, et al. Bioinformatic analysis identifies potential key genes in the pathogenesis of Melanoma. *Front Oncol*. 2020;10:581985. <https://doi.org/10.3389/fonc.2020.581985>.
- Ji C, Li Y, Yang K, et al. Identification of four genes associated with cutaneous metastatic melanoma. *Open Med*. 2020;15(1):531–539. <https://doi.org/10.1515/med-2020-0190>.
- Schäfer H, Subbarayan K, Massa C, et al. Correlation of the tumor escape phenotype with loss of PRELP expression in melanoma. *J Transl Med*. 2023;21(1):643. <https://doi.org/10.1186/s12967-023-04476-x>.
- Miranda E, Vizoso F, Martín A, et al. Apolipoprotein D expression in cutaneous malignant melanoma. *J Surg Oncol*. 2003;83(2):99–105. <https://doi.org/10.1002/jso.10245>.
- Wang Z, Chen H, Sun L, et al. Uncovering the potential of APOD as a biomarker in gastric cancer: a retrospective and multi-center study. *Comput Struct Biotechnol J*. 2024;23:1051–1064. <https://doi.org/10.1016/j.csbj.2024.02.015>.
- Ciccone V, Terzuoli E, Ristori E, et al. ALDH1A1 overexpression in melanoma cells promotes tumor angiogenesis by activating the IL-8/Notch signaling cascade. *Int J Mol Med*. 2022;50(1). <https://doi.org/10.3892/ijmm.2022.5155>.
- Luo Y, Dallaglio K, Chen Y, et al. ALDH1A isozymes are markers of human melanoma stem cells and potential therapeutic targets. *Stem Cell*. 2012;30(10):2100–2113. <https://doi.org/10.1002/stem.1193>.
- Ciccone V, Simonis V, Del Gaudio C, et al. ALDH1A1 confers resistance to RAF/MEK inhibitors in melanoma cells by maintaining stemness phenotype and activating PI3K/AKT signaling. *Biochem Pharmacol*. 2024;224:116252. <https://doi.org/10.1016/j.bcp.2024.116252>.
- Luo L, Zhang W, Li Z. LncRNA HAGLR may aggravate melanoma malignancy via miR-4644/ASB11 pathway. *Mol Biotechnol*. 2023;65(10):1619–1631. <https://doi.org/10.1007/s12033-023-00672-8>.

51. Wang J, Gao H, Fu P, et al. Knockdown of lncRNA-HAGLR restrains the viability and motility of pancreatic cancer via miR-625-5p/TAF15 axis in vitro and in vivo. *Heliyon*. 2024;10(18):e37254. <https://doi.org/10.1016/j.heliyon.2024.e37254>.
52. Foster HM, Carle MN, Jira LR, Koh DW. TRPM2 channels: a potential therapeutic target in Melanoma? *Int J Mol Sci*. 2023;24(13). <https://doi.org/10.3390/ijms241310437>.
53. McKamey SG, Jira LR, Tweed CM, et al. Antagonism of the transient receptor potential melastatin-2 channel leads to targeted antitumor effects in primary human malignant melanoma cells. *Int J Oncol*. 2022;60(4). <https://doi.org/10.3892/ijo.2022.5333>.
54. Eftang LL, Klajic J, Kristensen VN, et al. GFRA3 promoter methylation may be associated with decreased postoperative survival in gastric cancer. *BMC Cancer*. 2016;16:225. <https://doi.org/10.1186/s12885-016-2247-8>.
55. Amirouchene-Angelozzi N, Frisch-Dit-Leitz E, Carita G, et al. The mTOR inhibitor Everolimus synergizes with the PI3K inhibitor GDC0941 to enhance anti-tumor efficacy in uveal melanoma. *Oncotarget*. 2016;7(17):23633–23646. <https://doi.org/10.18632/oncotarget.8054>.